T-Rex

A Large-Scale Alignment of Natural Language with Knowledge Base Triples

In one line: T-REx is a giant, ready-made “answer key” that connects millions of Wikipedia sentences to the exact facts (triples) they are talking about in Wikidata — built automatically, not by hand.

Paper	T-REx: A Large Scale Alignment of Natural Language with KB Triples
Authors	Elsahar, Vougiouklis, Remaci, Gravier, Hare, Simperl, Laforest
Venue	LREC 2018
Size	3.09M Wikipedia abstracts ↔ 11M Wikidata triples
License	Creative Commons BY-SA 4.0 (data) / MIT (code)

1. What Problem Is This Paper Solving?

Imagine you want to teach a computer to read a sentence like “David Bowie was an English singer” and automatically understand: *this sentence is stating the fact (David Bowie, occupation, singer)*. To teach a model this skill, you need thousands of examples where a sentence is already matched to the correct fact. Creating these examples by hand is slow and expensive.

Before this paper, the datasets that already did this matching (pairing sentences with facts) were small, covered very few types of facts (called *predicates*, e.g. “born in”, “occupation”, “capital of”), or nobody had properly checked how accurate they were. The table below shows how weak the older resources were:

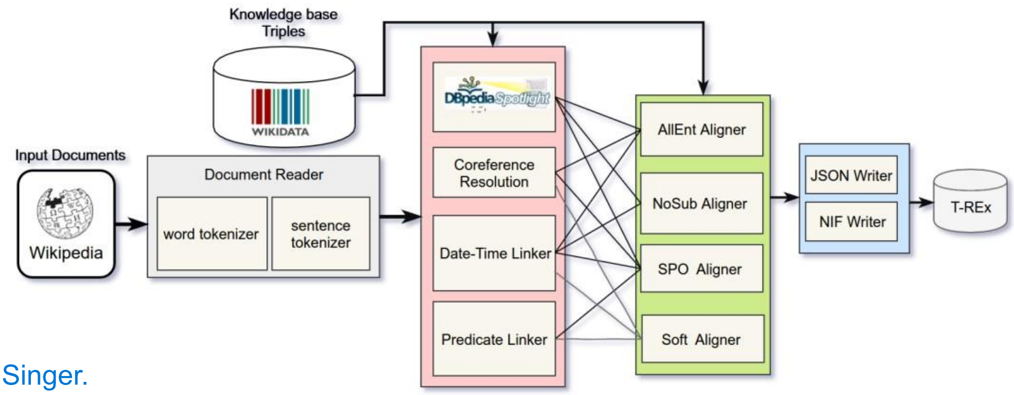
Dataset	Documents	Unique Predicates	Aligned Facts	Access
NYT-FB	1.8M sentences	258	39K	Partial
TAC-KBP	90K sentences	41	122K	Closed
Google-RE	60K sentences	5	60K	Public
FB15K-237	2.7M patterns	237	2.7M	Public
WikiReadings	4.7M articles	884	n/a (no real alignment)	Public
T-REx (this paper)	3.09M abstracts	600+	11M	Public

Table 1: How T-REx compares with earlier datasets that tried to do the same job.

In plain words: earlier datasets were either too small, only understood a handful of relationship types, or (like WikiReadings) didn't actually confirm the fact was mentioned in the text at all. T-REx is bigger and broader on every one of these measures — **about 100x bigger and 2.5x more diverse in predicates** than the best prior resource.

2. How T-REx Is Built — The Pipeline (Figure 1)

Rather than asking humans to label sentences, the authors built an automatic assembly line (a "pipeline") that takes in raw Wikipedia text and Wikidata facts, and outputs matched sentence-fact pairs. Here is the actual diagram from the paper, followed by a plain-English walkthrough of every box in it.



Ex:
David was an English Singer.
He was born in London.

Figure 1: Overview of the alignment pipeline and its components

Figure 1: Overview of the alignment pipeline and its components (as given in the paper).

Walking through the diagram, step by step:

Pipeline Step	Easy Explanation
Input (Wikipedia Documents)	The process starts with Wikipedia articles. These articles provide the text from which facts will be extracted.
Document Reader	Splits the article into words and sentences so the computer can understand the text easily.
Knowledge Base (Wikidata Triples)	Loads facts from Wikidata in the form of **(Subject, Predicate, Object)**. Example: **(David Bowie, occupation, singer)**.
DBpedia Spotlight (Entity Extraction)	Finds important names like people, places, and organizations in each sentence and links them to Wikidata.
Coreference Resolution	Replaces pronouns like **he, she, it** with the correct person's or object's name. He= David
Date-Time Linker	Finds dates and converts them into a standard format. Example: **8 January 1947** → birth date. -> 1947-01-08
Predicate Linker	Finds relationship words such as **"born in"** or **"actor"** and matches them to the correct Wikidata relation. Which relation is being expressed.
Triple Aligners (AllEnt, NoSub, SPO, Soft)	Checks whether the sentence really expresses a Wikidata fact and chooses the correct triple.

T-REx (Final Dataset)	The end result: a large, downloadable dataset of sentence ↔ fact pairs, ready to train models for tasks like Relation Extraction, Question Answering, or generating text from data.

3. Where the Raw Data Comes From

The text side of T-REx comes from the **DBpedia Abstracts** corpus — the English section alone has 4.6 million cleaned Wikipedia documents. The fact side comes from the **Wikidata 'truthy' dump**, which contains 144 million triples. The pipeline runs across all of this and keeps only the sentence-fact matches it is confident about, producing the final 3.09M document / 11M triple dataset.

4. A Full Worked Example

The paper uses this example sentence to show exactly how the three aligners behave differently:

> “David Bowie was an English singer, who later on **worked as** an actor. He was **born in** Brixton, London to his **mother** Margaret Mary and his **father** Haywood Stenton.”

Here is every candidate fact hiding in that sentence, and which aligner catches each one (a tick ✓ means that method found it):

#	Candidate Fact (Triple)	NoSub	AllEnt	SPO
1	David Bowie — nationality → England	✓		✓
2	David Bowie — occupation → singer	✓		✓
3	David Bowie — occupation → actor	✓	✓	✓
4	David Bowie — birthplace → Brixton	✓		✓
5	Brixton — region → London		✓	
6	David Bowie — child of → Margaret Mary	✓	✓	✓
7	David Bowie — child of → Haywood Stenton	✓	✓	✓
8	Margaret Mary — divorced → Haywood Stenton (wrong!)		✓	
9	Margaret Mary — death place → London (wrong!)		✓	

Table 2: Same sentence, three different aligners, three different sets of results.

Notice facts #8 and #9: **AllEnt** grabs them just because “Margaret Mary” and “London” both appear somewhere near each other in the text — but the sentence never actually says either of those things. This is exactly the trade-off the paper is highlighting: AllEnt catches the most facts, but also invents connections that aren't really there.

AlEnt- AI based entity alignment(matching entities and relation when wording is more complex)

5. The Three (Really Four) Aligner Strategies

1. NoSub Aligner

- Assumes every sentence in a paragraph has the same main subject.
- Does not require the subject to be explicitly written.
- **Strength:** Highest accuracy (98%) with very large coverage.
- **Weakness:** Cannot identify the exact position of the subject in the sentence.

2. AllEnt Aligner

- Connects every possible pair of entities in a sentence.
- Finds the maximum number of facts (about **11.1 million**).
- **Strength:** Captures almost every possible fact.
- **Weakness:** Produces the most noise because many linked entities are not actually related.

3. SPO Aligner

- Accepts a fact only when the **Subject, Predicate, and Object** are all explicitly present.
- **Strength:** Very high accuracy (96%) and clearly identifies each part of the fact.
- **Weakness:** Creates the smallest dataset (about **1.2 million** facts) because it rejects many valid cases.

4. Soft Aligner

- A more flexible version of the SPO aligner.
- Relaxes the strict SPO matching rules.
- **Strength:** Balances the precision of SPO with the wider coverage of AllEnt.
- **Weakness:** Not the main focus of the paper's evaluation, so fewer results are discussed.

6. Results — How Big and How Accurate?

Method	Accuracy	Total Alignments	Unique Predicates Covered
AllEnt	88%	11.1M	633
SPO	96%	1.2M	336
NoSub	98%	5.2M	642

Table 3: Size vs. accuracy trade-off across the three main aligners.

In simple terms: if you want the **most** data and don't mind some noise, use AllEnt. If you want the **most trustworthy** data with good coverage, use NoSub. If you need to know exactly **where** a fact sits in a sentence (e.g. for extractive QA), use SPO.

Accuracy on some individual predicates (Table 5 in the paper) also shows an interesting pattern:

Predicate	AllEnt	SPO	NoSub
date of birth	1.00	1.00	1.00
country of citizenship	0.91	1.00	0.95
occupation	0.90	0.94	1.00
spouse	0.75	0.94	1.00
capital	0.40	1.00	n/a
shares border with	0.14	1.00	n/a

7. How Did They Check the Quality? (Evaluation Process)

Numbers like “98% accuracy” don't mean anything unless we know how they were measured. Here's what the authors actually did:

- They picked 2,600 aligned sentence-fact pairs from 700 Wikipedia abstracts, chosen so this sample matched the overall dataset's statistics (not biased toward any one topic).
- Real human reviewers (via crowdsourcing) read each sentence and judged: 'Is this fact actually and explicitly stated in this sentence?' — not just implied or guessed.
- Every single alignment was checked by at least 5 different people, and the majority vote decided if it counted as correct.
- To keep quality high, the authors first hand-labelled 100 documents themselves and used those as 'test questions' hidden among the real tasks. Any reviewer scoring below 80% on these hidden tests was removed from the whole experiment (their answers were thrown out as unreliable).
- They also calculated an **Inter-Annotator Agreement** score — basically, 'how much did the 5 reviewers agree with each other?' High agreement means the judgement was clear-cut, not guesswork.

Method	Accuracy	Inter-Annotator Agreement
AllEnt	88%	0.85
SPO	96%	0.93
NoSub	98%	0.96

8. What Actually Goes Wrong? (Error Analysis)

The authors manually looked through a sample of incorrect alignments and found three repeat offenders:

Error Types (Easy Study Notes)

1. Nested Relations

- **Real example:** "He was the son of Ekoji I and the younger brother of Serfoji I."
- **Wrong output:** *(Ekoji I, child, Serfoji I)*
- **Why it happens:** Two different relationships (son and brother) appear in the same sentence, confusing the aligner.

2. Wrong Entailment

- **Real example:** "Ernst Gustav Kuhnert was born in Tallinn, Estonia."
- **Wrong output:** *(Tallinn, capital of, Estonia)*
- **Why it happens:** The model assumes a relationship because the entities appear together, even though the sentence never states it.

3. Entity Linking Errors

- **Real example:** "Carolyn Virginia Wood is an American..."
- **Wrong output:** *(Virginia, country, American)*

- **Why it happens:** The entity linker mistakes "Virginia" (part of the person's name) for the place "Virginia".

9. Limitations

- AllEnt struggles badly on relations like 'capital' or 'shares border with', where two places being mentioned together rarely means the relationship is true.
- The whole system depends on automatic entity linking (DBpedia Spotlight) — if that makes a mistake, the error carries through the rest of the pipeline.
- Nested relations inside short, complex sentences are hard to separate correctly.
- The pipeline only looks at one sentence at a time — it can't reason across multiple sentences or the whole document.

10. Availability and Licensing

T-REx is free and public. The dataset itself is released under a **Creative Commons Attribution-ShareAlike 4.0** license, and the pipeline's source code is released separately under the **MIT License** — meaning anyone can use, adapt, and build on both, as long as they give credit and (for the data) share alike.

11. Conclusion and Future Work

T-REx set a new bar for datasets that connect free text to structured knowledge: about 100x bigger and 2.5x more predicate-diverse than the best resource before it, with a measured 97.8% overall accuracy confirmed by careful human evaluation. Because it offers three (really four) aligners with different coverage/precision trade-offs, downstream researchers can pick whichever fits their task — from training high-coverage models to building precise extractive QA systems.

The authors point to three directions for future improvement: (1) using sentence grammar/dependency structure to fix nested-relation errors, (2) adding logical implication rules to catch wrong-entailment errors, and (3) improving the underlying entity linker to reduce linking mistakes.

Source: Elsahar, H., Vougiouklis, P., Remaci, A., Gravier, C., Hare, J., Simperl, E., & Laforest, F. (2018). T-REx: A Large Scale Alignment of Natural Language with Knowledge Base Triples. LREC 2018.